concentrations such as mouthwash, antiseptic solutions, household disinfectants etc... Have you ever noticed the concentration of the ingredients present in those solutions? For example, chlorhexidine mouthwash solution contains 0.2 % (w/v) chlorhexidine gluconate; The concentration of the commercially available hydrogen peroxide is 3% (w/v). Similarly, other terms such as ppm (TDS of water), molar and normal (laboratory reagents) are used to express the concentration of the solution. The concentration of a solution gives the amount of solute present in a given quantity of solvent. As we have seen, there are different ways of expressing the concentration of a solution. Let us learn the different concentration terms and to prepare a solution of a specific concentration.

Table 9.2 Different concentration units and their illustrations

S. No.	Concentration term	Expression	Illustration
1	Molality (m)	$\frac{\text{Number of moles of solute}}{\text{Mass of the solvent (in kg)}}$	The molality of the solution containing 45 g of glucose dissolved in 2 kg of water $\frac{\text{Number of moles of solute}}{\text{Mass of the solvent (in kg)}} = \frac{\left(\frac{45}{180}\right)}{2}$ $= \frac{0.25}{2} = 0.125 \text{ m}$
2	Molarity (M)	$\frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}}$	5.845 g of sodium chloride is dissolved in water and the solution was made up to 500 mL using a standard flask. The strength of the solution in molarity is $\frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}} = \frac{\left(\frac{5.845}{58.45}\right)}{0.5}$ $\frac{0.1}{0.5} = 0.2 \text{ M}$
3	Normality (N)	$\frac{\text{Number of gram equivalents of solute}}{\text{Volume of solution (in L)}}$	3.15 g of oxalic acid dihydrate, is dissolved in water and the solution was made up to 100 mL using a standard flask. The strength of the solution in normality is $= \frac{\text{Number of gram equivalents of solute}}{\text{Volume of solution (in L)}}$ $= \frac{\left(\frac{\text{mass of oxalic acid}}{\text{Equivalent mass of oxalic acid}}\right)}{\text{volume of solution (inL)}} = \frac{\left(\frac{3.15}{63}\right)}{0.1}$ $= \frac{0.05}{0.1} = 0.5 \text{ N}$

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